

Dataset Reference Document

7 Groups, 8 Datasets, 1 Unified Project Framework

Data Mining Course

Overview

Each group is assigned a distinct banking-domain dataset. All groups execute the same 5-phase project. Datasets differ in domain focus, scale, and mining characteristics, ensuring groups cannot share results directly while enabling meaningful cross-group comparison at the Mining Expo.

All datasets are publicly available and free to download. Datasets marked with Kaggle Login require a free Kaggle account. No paid subscriptions are required for any group.

Master Dataset Assignment Table

Grp	Dataset	Records	Feat.	Access	Best Phase
1	UCI Bank Marketing	41,000	20	Free, No Login	Association Rule Mining
2	Lending Club Loans	890,000	74	Free, Kaggle Login	Clustering
3	PaySim Fraud Transactions	6,300,000+	11	Free, Kaggle Login	Anomaly Detection, ARM
4	HMDA Mortgage	Millions (100K)	99	Free, No Login	ARM, Visualization
5	Bank Customer Churn	10,000	14	Free, Kaggle Login	Clustering, ARM
6	Banking Transactions	1,000,000+	9	Free, No Login	Pattern Mining
7	FDIC Bank Financial Data	100,000+ filings	100+	Free, No Login	Outlier Detection
8	IBM AML Transactions	5,000,000+	11	Free, Kaggle Login	Anomaly, ARM

Group Dataset Details

Group 1: UCI Bank Marketing Dataset

Domain Focus	Customer response to term deposit marketing campaigns
Records	41,188 records

Features	20 features
Source	UCI Machine Learning Repository
Link	archive.ics.uci.edu/dataset/222/bank+marketing
Login Required	No, fully free, direct CSV download
Mining Angle	Discover what customer profiles respond positively to campaigns. Mine rules such as: retired customers with high balances who were not previously contacted are likely to subscribe to term deposits.

Group 2: Lending Club Loan Dataset

Domain Focus	Loan issuance, risk grading, and borrower financial characteristics
Records	~890,000 records
Features	74 features
Source	Kaggle (Lending Club historical data)
Link	kaggle.com/datasets/wordsforthewise/lending-club
Login Required	Yes, free Kaggle account required
Mining Angle	Cluster borrowers by risk profile using grade, income, and debt ratio. Mine rules such as: borrowers stating small business as loan purpose with 10 or more years of employment tend to receive Grade A loans with low interest rates.
Note	74 features require serious Phase 1 feature reduction before clustering. Students must select a meaningful analytical subset early.

Group 3: PaySim Synthetic Financial Fraud Dataset

Domain Focus	Mobile money transaction fraud detection
Records	~6,300,000 records
Features	11 features
Source	Kaggle, Lopez-Rojas et al. (2016)
Link	kaggle.com/datasets/ealaxi/paysim1
Login Required	Yes, free Kaggle account required
Mining Angle	Mine frequent transaction patterns across transaction types (CASH-OUT, TRANSFER, PAYMENT). Cluster accounts by flow behavior using amount ranges and balance deltas. Detect layering anomalies through unusual balance changes and round-trip transfer patterns.

Note	The fraud label exists in the dataset but must not be used as a classification target. It may only be consulted after mining to validate whether discovered anomalies align with actual fraud cases.
-------------	--

Group 4: Home Mortgage Disclosure Act (HMDA)

Domain Focus	U.S. mortgage application outcomes across demographics, geographies, and institutions
Records	Millions of records, use a 100,000-record sample
Features	99 features
Source	Consumer Financial Protection Bureau (CFPB), U.S. Government Open Data
Link	ffiec.cfpb.gov/data-publication/snapshot-national-loan-level-dataset/2022
Login Required	No, fully free U.S. government open data, direct CSV download
Mining Angle	Mine approval and denial patterns across income bands, demographics, and property types. Segment applicants by financial profile and location. Discover rules on which applicant combinations are systematically approved or denied.
Note	Students must sample 100,000 records from the full dataset before beginning Phase 1 to keep processing manageable.

Group 5: Bank Customer Churn Dataset

Domain Focus	Identifying disengaging customers in retail banking across multiple countries
Records	10,000 records
Features	14 features
Source	Kaggle
Link	kaggle.com/datasets/shubhammeshram579/bank-customer-churn-prediction
Login Required	Yes, free Kaggle account required
Mining Angle	Cluster loyal versus disengaged customers by behavioral and financial attributes. Mine rules such as: customers from Germany holding only one product who are inactive represent a strong churn profile. Detect anomalies in sudden balance drops preceding account closure.

Group 6: Banking Transaction Dataset

Domain Focus	Day-to-day banking transaction behavior and customer demographics
Records	1,000,000+ records
Features	9 features

Source	GitHub, Open MIT License (ahsan084/Banking-Dataset)
Link	github.com/ahsan084/Banking-Dataset
Login Required	No, fully free, direct GitHub download
Mining Angle	Segment customers by transaction frequency and type across account categories. Mine rules on withdrawal and deposit patterns by age group, gender, and account type. Detect branch-level anomalies in unusual transaction volume or amount spikes.

Group 7: FDIC Bank Financial Data (Call Reports)

Domain Focus	Institutional-level financial health and stability of U.S. commercial banks
Records	100,000+ quarterly filings
Features	100+ features
Source	Federal Deposit Insurance Corporation (FDIC), U.S. Government Open Data
Link	banks.data.fdic.gov/docs
Login Required	No, fully free, API and CSV download available
Mining Angle	Cluster banks by financial health profile using capital ratios, profitability, and asset quality. Mine rules linking bank size to profitability indicators. Detect anomalous institutions with unusual capital adequacy ratios or non-performing loan spikes.

Group 8: IBM Synthetic AML Transactions

Domain Focus	Money laundering pattern discovery in large-scale transaction networks
Records	5,000,000+ records
Features	11 features
Source	IBM Research, Kaggle
Link	kaggle.com/datasets/ealtman2019/ibm-transactions-for-anti-money-laundering-aml
Login Required	Yes, free Kaggle account required
Mining Angle	Mine frequent transaction patterns across currency types and payment formats. Cluster accounts by transaction flow behavior. Detect layering anomalies using outlier analysis on transaction chains and round-trip flows.
Note	The laundering label must not be used as a classification target. It may only be consulted after mining to validate discovered anomalies.

Important Notes for All Groups

1. Groups 3 and 8 contain fraud or laundering labels. These must not be used as classification targets during any phase. They may only be consulted after mining is complete to validate findings.
2. Group 4 (HMDA) must sample 100,000 records from the full dataset before beginning Phase 1 to keep processing manageable.
3. Group 2 (Lending Club) has 74 features. Phase 1 feature selection is critical. Students must reduce to a meaningful analytical subset before clustering.
4. All datasets are in CSV or API format and are compatible with Python (pandas, scikit-learn, mlxtend) and standard data mining tools.